

Ask Data - Complete Agentic AI Project Guide

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This report explains the full agentic setup in your project, what each agent does for users, what each agent does in the backend pipeline, and how modules are separated in UI.

1. Is This Now an Agentic AI Project?

Yes. This is now a bounded multi-agent analytics system. It does not use one uncontrolled mega-agent. Instead, the flow is split into smaller agents with clear responsibilities: intent routing, planning, governance checks, generation, repair, execution, insight, and trend watch.

Current maturity: practical production-oriented agentic foundation with read-only governance and traceable execution.

2. Agentic Architecture in One View

- User Prompt
- > Intent Router Agent
 - > (Direct Answer OR Clarification OR Data Query Path)
 - > Query Planner Agent
 - > Governance Policy Agent
 - > SQL Generation Agent
 - > SQL Repair Agent (if needed)
 - > Multi-source Execution Agent
 - > Business Insight Agent
 - > Trend Watch Agent
 - > Agentic Trace + UI Panels

3. What Each Agent Does

Agent	Role In Project	What User Gets	Guardrail
Intent Router Agent	Classifies prompt: direct answer, metadata question, profile/context/schema/table discovery, or SQL path	Direct answer for profile/context/schema/table discovery	Only questions SQL path when necessary
Query Planner Agent	Selects best connection/tables and columns	Best source targeting and fewer wildcards	Can trigger clarification when confidence is low
SQL Generation Agent	Builds safe query from intent + schema	Natural language to data output	Read-only constraints before execution
SQL Repair Agent	Retries failed SQL using error-aware hints	Higher success on first-run failures	Bounded retry count
Governance Policy Agent	Enforces RBAC, row caps, query/expansion limits	Enterprise-safe behavior and permissions	Strict lookback relations
Execution Agent	Runs across allowed SQL/Mongo data sources	A unified result set across selected sources	Connection-level permission checks
Business Insight Agent	Converts raw rows to business narrative	Readable business summary instead of only raw data	Suppresses identifier-like noise fields
Trend Watch Agent	Computes week-over-week change and trends	Proactive signals/anomaly detection	Only emits trend when time axis is valid

4. What Users Experience

- Natural-language answers for organisation/profile/context questions.
- Schema/table/column discovery answers without forcing SQL every time.
- Business summary in plain language after data execution.
- Trend Watch outputs for week-over-week movement.
- Proactive alerts for anomalies such as spikes, drops, or null patterns.
- Clarification prompts when the request is ambiguous.

5. UI Module Separation (Latest)

- Natural Bot is now a standalone top-level module.
- Analytics Studio remains a separate module for generated queries, results, charts, and trace inspection.

- Data Sources and Schema modules stay independent for governance and discoverability.
- This separation prevents mixing chat intent collection with studio analysis review.

6. Why Agentic Panels Can Look Empty

- Analysis request failed (auth, permission, backend down, bad query).
- Prompt is ambiguous and agent asks for clarification instead of execution.
- No rows returned, so trace has limited sections.
- Backend not running or blocked by task startup command issue.

The UI displays current error details when trace data is missing, so users can identify the cause quickly.

7. Role-Based Agentic Behavior

Role	Primary Agentic Value
Employee	Ask natural questions safely within assigned permissions; receive trends and summaries.
Admin	Manage data/permissions and run agentic analytics with governance visibility.
Super Admin	Cross-organisation governance and platform-level control (foundation ready).

8. Runtime Note

In this workspace, the predefined backend task command is misquoted around a path with spaces ('c:\my projects...') and fails with 'c:\my is not recognized'. Backend endpoint can still be up if another process already occupies port 8000.

9. Next Agents to Add

- Connection Triage Agent (super admin): classify auth/network/db/schema failures with confidence and evidence.
- Data Quality Watchdog Agent: scheduled checks for freshness/null spikes/row-count drift with suppression windows.
- Permission Recommendation Agent (admin): least-privilege suggestions requiring human approval.
- Report Builder Agent: convert goal to scheduled report with preview + approval.

10. Agentic Sequence (Current)

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Prompt -> Planner score -> (Clarification if low confidence)
      -> Policy & RBAC checks -> SQL generation
      -> Repair loop (max retries) -> Execution
      -> Business analytics cleanup -> Trend snapshot
      -> Proactive alerts + human brief -> UI trace
```

Conclusion

Your project is now agentic in a practical, enterprise-safe way: separate agents, clear guardrails, transparent traces, and role-aware user outcomes.